

Middleware for heterogeneous subsystems interoperability in intelligent buildings

Thinagaran Perumal^{*}, Abd Rahman Ramli, Chui Yew Leong, Khairulmizam Samsudin, Shattri Mansor

Intelligent Systems and Robotics Laboratory, Institute of Advanced Technology, Universiti Putra Malaysia, 43400 Serdang, Selangor Darul Ehsan, Malaysia

ARTICLE INFO

Article history:
Accepted 12 November 2009

Keywords:
SOAP
Intelligent building
ECA
Web Services
Interoperability
Middleware

ABSTRACT

Managing heterogeneous subsystems in an intelligent building has always been elusive due to lack of interoperability factor. This is due to the nature of heterogeneous systems in an intelligent building as a distributed entity that needs a certain degree of interoperability and joint execution of tasks for managing heterogeneous systems that consist of different platforms and protocols. There is a need for interoperability in an intelligent building with a bespoke solution that could make the heterogeneous subsystems 'talk' with each other and interoperate regardless of the operating platform. To solve the interoperability problem, Web Services technology is seen as a potential solution that could deal with the dynamic environment of an intelligent building. In this paper, we describe a SOAP-based middleware implementation for managing heterogeneous subsystems in an intelligent building. The proposed middleware solution utilizes the SOAP technology together with ECA rule mechanism in managing subsystems interoperation in a federated manner. Experiments and performance evaluation of the middleware had been carried out to justify the performance requirement of heterogeneous systems in an intelligent building.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

In recent years, there are many emerging trends that drive the quest for subsystems interoperability in intelligent buildings. An intelligent building is defined as an entity that creates an environment that maximizes efficiency of building services and at same time ensures effective resource management with minimum life-time costs [1]. Intelligent buildings are often characterized by integration of heterogeneous subsystems in a unified manner. Heterogeneous subsystems like HVAC&R, digital surveillance system, access control and fire management system are some significant infrastructures that shape the intelligent building environment [2]. Fig. 1 below shows the heterogeneous systems in a typical intelligent building environment.

In a broader sense, an intelligent building is concerned with improving interoperability among its components in an intelligent manner. As such, many recent works provide solutions with many facets to cater cohesive integration between building subsystems [3,4]. There are also several solutions specifically targeting intelligent integration which have come under many forms including the use of artificial intelligence to make buildings more intelligent [5,6]. In general, many of these current subsystems use proprietary information patterns and communication protocols that prevent the easy modification of the intelligent building operation, and at the same time limit the plug and play applications and addition of new components in the system. Such implication leads to interoperability

issues among subsystems. Although significant progress has been achieved in intelligent building integration research, the interoperability issue is still elusive when it comes to heterogeneous subsystems integration [7]. In recent years, it is evident that the combination of low cost, high performance embedded systems together with the emergence of high capacity communication infrastructure and the Internet has dramatically led to the explosive growth of subsystems in intelligent buildings. The growth of such subsystems has been gradually transforming intelligent buildings into a data-intensive environment with many subsystems and front-end devices configured for specific application domains. This transformation results in a few challenges, particularly associated with the management of subsystems in an intelligent building. The first challenge observed in an intelligent building scenario is the difference of resources, protocols, operating platforms and programming languages utilized among the heterogeneous subsystems configured. Such diversified platform with multiple subsystems in the building environment promotes the growth of gateways. Many system integrators and manufacturers opt for the use of gateways at the building server infrastructure to tie together subsystems from different manufacturers. Gateways are always seen as a possible solution for interoperability. However, these gateways do not seem to provide full functionality that guarantees multiple subsystems integration. A typical gateway only allows limited sources of information like system status and control messages to be exchanged between different subsystems. Critical information like subsystem fault status could not be shared and data from different subsystems wasn't accessible for joint event execution [8]. Furthermore, the gateways needed to be reconfigured especially when one of the subsystems was modified or

^{*} Corresponding author. Tel.: +60 126594933; fax: +60 386566061.
E-mail address: thinagaran@hotmail.com (T. Perumal).

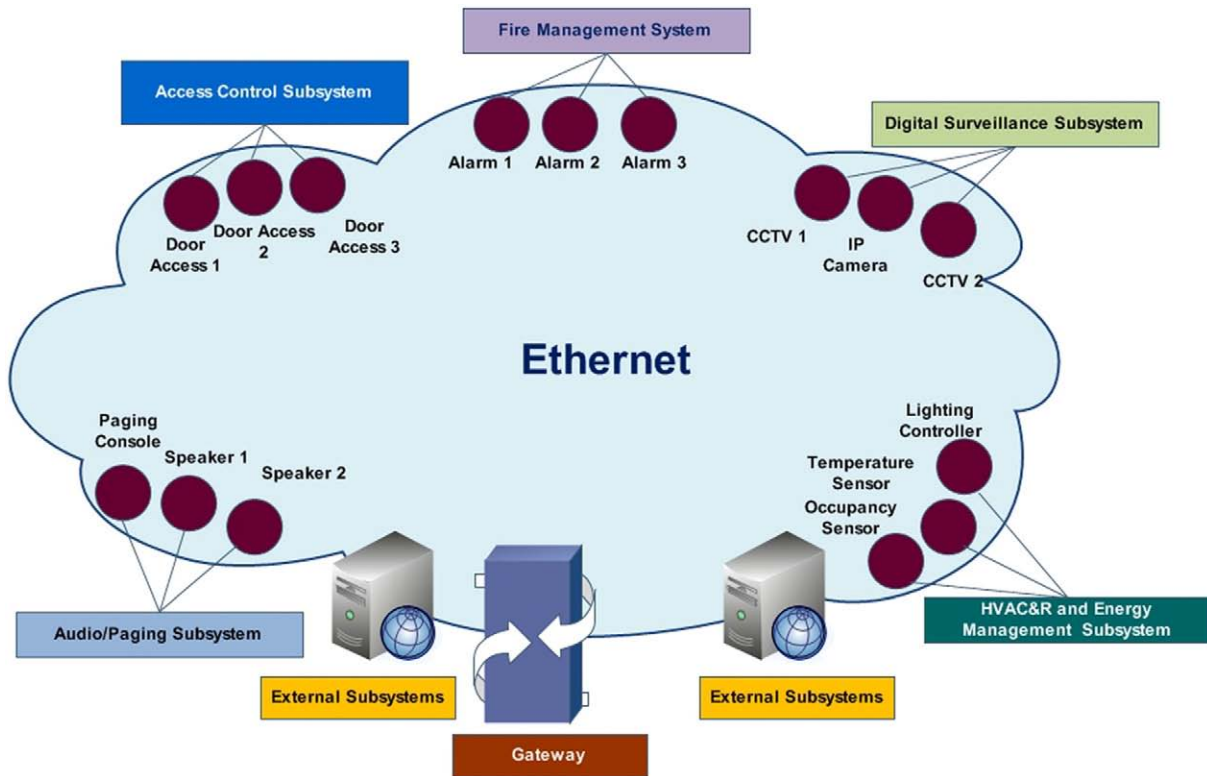


Fig. 1. Heterogeneous systems in an intelligent building.

added, creating troubleshooting issues and support problems. These consequences lead to additional expenses and maintenance cost for facility managers or building owners. The second challenge is the coordination of those heterogeneous subsystems in the intelligent building. Various subsystems need to perform interoperation to realize particular building operation, thus a federated coordination is needed to ensure timely data exchange among them.

In recent times, it is noticeable that XML and Web Services technologies are seen as a potential answer in providing a degree of interoperability for the intelligent building [11]. During the past years, XML and Web Services have been increasingly seen as a solution for a larger distributed system [12]. These criteria fit the intelligent building due to its diversification subsystems with various applications. The fundamental entities of Web Services are *publish*, *discover* and *invoke*. These fundamental entities are defined by Web Services standards known as SOAP which was originally defined as Simple Object Access Protocol, Web Services Description Language (WSDL) and Universal Description, Discovery and Integration (UDDI) [13]. Among these standards, SOAP acts as a *glue* that lets users and service providers to ‘talk’ with each other and exchange XML data to enable communication across the Web [11]. SOAP defines an XML envelope to carry messages and invoke interaction between systems in a standard way that is operating system, programming language and geographically independent. Using SOAP technology, Web Services are invoked by initiating a service invocation into XML message format. Recently, Web Services also gained attention from application developers when Open Building Information eXchange (OBiX) initiated by OASIS creates a comprehensive standard for information exchange among various subsystems that could facilitate home and building industries [14]. In this paper we will discuss the implementation of SOAP-based middleware for interoperability among heterogeneous subsystems in the intelligent building. The concern is to solve two important issues; integration using XML SOAP and joint execution of tasks between heterogeneous subsystems in the intelligent building. The paper is organized as follows. Section 2

introduces the issues and constraints in the current intelligent building integration and their related work. Section 3 presents the SOAP-based middleware architecture and its implementation. Section 4 presents the experimental design and performance evaluation. Finally, the conclusions and discussions are presented in Section 5.

2. Background and related work

To develop a middleware that caters full interoperability features requires many factors to be considered. Some important factors that require consideration are scalability, accessibility and reliability. For an intelligent building, scalability is important to ensure new subsystems to be added as well as new dependencies without much modification on the entire system. The designed middleware must allow the subsystems to be expanded without any service interruption and system modification. As new devices and subsystems are increasingly introduced into intelligent building, a middleware with the flexibility to add a new subsystem is needed whenever a modification or system configuration is required. Middleware for an intelligent building should also provide easy accessibility for all data type transactions performed by heterogeneous subsystems. The debate over a universal platform for intelligent building has been never-ending, as the demand for interoperable systems, providing increased service, faster response, and greater flexibility, has escalated. The need for interoperability arises because in order for an intelligent building to operate with greater efficiency and cost savings, data must be shared. Most of these systems employ proprietary protocols, owned by individual vendors or design specialists that are not available to the public, thus preventing data sharing and message exchange between different systems. With proprietary protocols, there is no “mix and match” of various systems and technologies. Hence, the need for interoperability is seen as a must to cater the evolving nature of intelligent building. IEEE defined interoperability as two or more dissimilar systems that perform joint execution of tasks [9]. In the view of an intelligent building,

interoperability generally involves the intercommunication and interaction between two or more subsystems to achieve comfortable living more effectively than a similar system would do independently. In the intelligent building perspective, interoperability defines the process by which subsystems from different manufacturers exchange information with the absence gateways, protocol translators, or other networking devices [10]. Achieving interoperability requires a common way of communicating between the different subsystems as well as managing, commissioning and maintaining them. The benefits delivered by interoperability are many. For instance, one sensor or building device can be shared among many different subsystems, allowing fewer sensors or building devices to be deployed and the overall cost of the building system drops appreciably. These subsystems include heating, ventilation, air conditioning and refrigeration (HVAC&R), lighting control, digital video surveillance, audio paging and evacuation system, fire management systems and so forth. Besides working on common data format, subsystems from different vendors also need a certain degree of intelligence to perform their joint task execution accordingly. An intelligent interaction that corresponds for decision making among subsystems is needed to support multiple service integration in an intelligent building. Many of existing legacy systems are interfaced with dry-contact form. There is a huge consideration towards data-oriented interface rather than a dry-contact form for those evolving legacy subsystems. Data-oriented interface seems more suitable to support contemporary demands of services in an intelligent building. For example, in considering interoperation between a digital video surveillance subsystem and an audio paging subsystem, using the dry-contact approach requires a facility manager to page manually to the target zone that triggers a particular event with motion detection. An automated decision making is required to ensure a paging message streamed to the target zone where the event of motion detection happened, and perform a selection of messages for working hours as well as warning messages for the time period after working hours. Therefore, to ensure full interoperation among heterogeneous subsystems, efficient decision making elements have to be included as part of the building operation mechanism. This is to ensure that heterogeneous subsystems can be interoperating intelligently with each other in order to perform joint execution of tasks. During such interoperation, an event occurring in one subsystem could invoke a series of triggering responses in other systems. For example, if the fire management systems detect a fire, it would provide a coded alarm message that would alert the HVAC&R system to execute smoke containment via zone pressurization. In the same scenario, the fire management system can cast the alarm message containing fire zone location to the digital video surveillance system and lighting control system, switching exit doors from a closed mode to access open mode and pulsing the lighting systems situated along the corridors leading to an exit pathway. The facility manager can monitor the affected zone situation remotely via the digital surveillance systems and alternatively instruct evacuation procedure manually or autonomously. Bringing all subsystems from different vendors and realizing the desired operation in an intelligent building is indeed a tremendous effort. For example, integrating fire alarm management systems message with digital surveillance systems streaming can be challenging enough due to their high demands made on their dependability.

Hence, deploying interoperability in an intelligent building could increase comfort of building residents at a reduced operational cost. Costs reduction are also possible by deploying access to all building services system in an interoperable, centralized monitoring and control centre. Having a centralized monitoring and control center allows faulty conditions to be detected and fixed at an early stage with minimum effort. This is especially applicable when a subsystem is controlled from a remote location. A unified platform for all subsystems further eases the task of the facility manager. In addition, such platform would allow hybrid integration with

enterprise network (local LAN/WAN) for management tasks such as log report generation, cost allocation and accounting. Facility managers could benefit in terms of (life-cycle) cost and more importantly maximizing building functionality as more systems are integrated together.

Currently, there are many legacy subsystems in dry-contact form that don't provide stored data accessibility and this proves to be a drawback when it comes to interoperation requirement. Therefore, a middleware for an intelligent building should provide easy accessibility to use all data types in an intelligent manner using functional rules or specifications. Presently, there many types of data associated in heterogeneous subsystems. Table 1 below shows the data types for heterogeneous subsystems.

These data types are generally discrete values with Boolean parameters "ON" or "OFF" in many subsystems. Some data types are in ranges like the one used for temperatures in HVAC&R subsystem. Many subsystems like fire management systems and digital surveillance subsystems are critical to an intelligent building environment. Data transmission between subsystems must be reliable and reach the destination in a timely manner and without significant delay. Existing subsystems are meant to operate in isolation and thus there is no platform for them to realize a joint operation. Each data type from particular subsystems contains critical information. Hence, the middleware designed needs to cater full reliability by ensuring data types of respective subsystems to reach the server in a federated manner and execute the designated tasks accordingly.

A number of articles in the literature have discussed several systems that implement interoperability among subsystems in an intelligent building. To solve integration and interoperation problem in an intelligent building environment, the middleware seems to be the ideal choice for both data representation and joint exaction of tasks. The middleware is a layer above the operating system and communication protocol layer but below the application layer that simplifies the communication between various systems [15]. By encompassing interaction between disparate systems and databases, the middleware enables unified user interfaces, reduces investment in infrastructure requirements, and allows heterogeneous subsystems to be easier to manage. One of the ideal solutions for intelligent building systems integration was proposed by Wang et al. [16]. Their work was focused on the design and implementation of a middleware technology based on Web Services combined with OPC to accommodate different communication interfaces in a building environment. Although detailed discussion was given in their work regarding the deployment of OPC interfaces wrapped as Web Services, there seems to be a lack of discussion on how interoperation takes place between heterogeneous systems. OPC uses stateful message exchange patterns and this means when an intelligent building operation gets disconnected, OPC might not be able to reinitialize the connection. This is crucial in intelligent building operation as all data types from each subsystem should be received without interruption to ensure reliability. OPC technology eliminates the requirement for developing customized interface for data exchange by providing a common interface that permits task execution repeatedly. OPC has really benefited the building automation and control industry and better

Table 1
Data types for heterogeneous subsystems.

Subsystems	Data types
Audio subsystems	Sensor data, dry contact status and audio channels
Digital surveillance subsystem	Video channel control data, pan/tilt/zoom (PTZ) control data
HVAC and energy management subsystem	Temperature, dry contact status
Fire management subsystem	Sensor data

potential of its features is expected with its new specification called OPC Unified Architecture [17]. Apart from OPC, several open solutions are also available in recent times. One such open solution in solving the management issues is using network gateways (consisting hardware and software modules) between defined building networks. These network gateways provide interface and translation mechanism between networks for managing disparate systems in the building environment. An example is the interface between UPnP and LonWorks as highlighted by Cemishkian et al. [18] in their work. However, this proposed method lacks scalability as the number of gateways increases dramatically with respect to the number of subsystems to be connected. Middleware standards like CORBA [19] and DCOM [20] are primarily for general Internet applications and they have some problems when directly used for the building management systems. Network messages from subsystems facing difficulties passing through firewalls make them not preferable for building systems. It is important to highlight that many subsystems in an intelligent building are time-critical especially security and fire management systems, therefore data of each event triggered by subsystems must be able to be exchanged without constraints. Another important solution comes from OSGi alliance [21]. OSGi is an open standard based software framework that combines bundles of services. Since OSGi is based on managed Java framework, the facility manager may need to download Java Virtual Machine (JVM) into the external subsystems or client terminal each time to perform control operation and management. Another similar solution that supports heterogeneity was presented by V. Kapsalis et al. using web service for various service provisions [22]. The work presented was intended for home environment which also seems to be applicable in the building scenario. The emphasis was given towards introducing various e-services with domains like healthcare and security. However the work highlighted the lack of performance test which is crucial for Web Services based implementation. A common feature of all these middleware proposed is that they focus on a single platform implementation (device level) and does not work heterogeneously. Their functionalities are isolated and not meant for interoperation. If a new subsystem needed to be integrated, a different architecture and software system has to be deployed. In many cases, the legacy subsystems and the newly deployed could not 'talk' with each other during integration. Another drawback is that all these subsystems are required to keep a separate network connection for their operation in an intelligent building. The number of separate connections will consume up to the number of gateways. In addition, each gateway will constitute as an entry point of each network in the building and this

eats up the bandwidth. Gateways also need to be equipped with better security enhancement or firewalls for full protection availability in building networks. Finally, each time new services are introduced, the configuration of the networks also changes and this leads to additional service or status update into the building owner's databases. Popular solutions like BACNet [23] and LonWorks [24] are much preferred for building automation and controls systems, and also accommodate Web Services interface as part of their standards to solve integration. Though common data format transportation could be achieved for intelligent building integration, full scale interoperation still seems to have a long way to go. In recent times, many facility managers and building owners realized the importance of integrating building applications with enterprise infrastructure like Enterprise Resource Planning and Management Information Systems. Such integration can be solved through Web Services. This new style of services has evolved with the means to provide direct interaction between dissimilar systems. The pervasive nature of the Web Services has encouraged the creation of integration techniques for distributed platforms. It is worth mentioning that the Continental Automated Buildings Association (CABA) has initiated a taskforce called CABA/XML Web Services Guideline Committee which takes care of the awareness and deployment procedures of XML and Web Services within building systems. The committee was later revamped to form a more specific niche organization called OBiX (OASIS Open Building Information eXchange Technical Committee). Using the Web Services technology, heterogeneous systems from different manufacturers on different operating platforms can be integrated easily. Fig. 2 below shows the Web Services as an enabler for interoperation among heterogeneous subsystems.

With Web Services, integration via common message format transportation can be realized to support interoperability among heterogeneous subsystems. However, there is a considerable need to represent the subsystems interoperation in an event-driven manner together with a common data format. XML SOAP is adopted to ensure maximum interoperability for an intelligent building. SOAP is an open standard Web Service protocol initiated by W3C (World Wide Web Consortium) [25], platform independent and compatible with mainstream programming languages such as .NET Framework, Java, PHP and so forth. This means that the middleware could support and interoperate with third party applications or subsystems running on heterogeneous systems platform. Using SOAP-based middleware powered by backend database, facility managers could easily bridge applications written with DCOM, CORBA or JAVA RMI as long as the service request is done using SOAP.

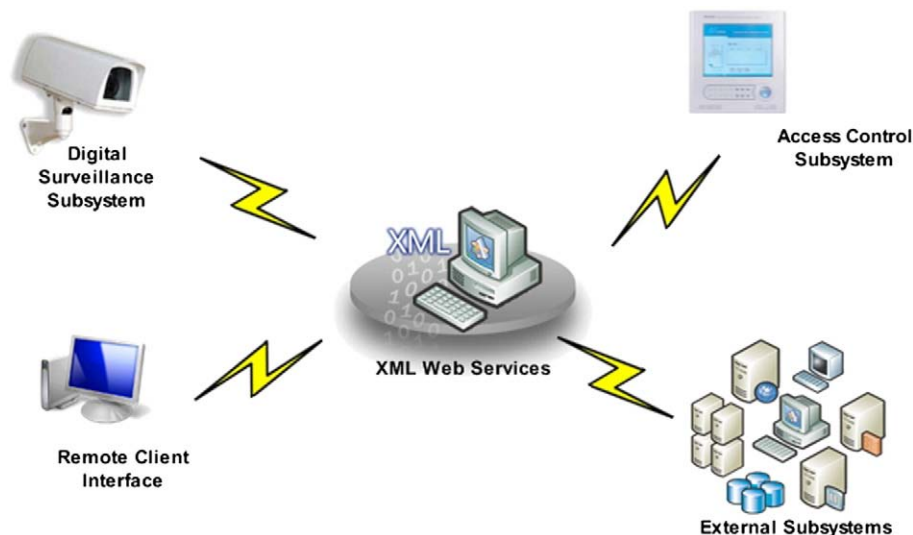


Fig. 2. XML Web Services for heterogeneous subsystems.

3. Implementation

Interoperability challenges arise when dealing with disparate data provided by heterogeneous subsystems. Every subsystem will produce its own event that corresponds to the dedicated functionalities in an intelligent building. For every event triggered by the subsystems, each message has to be encoded in a common format over the network. In addition, to ensure the data types provided by each subsystem are accessible and interoperable with a large range of services, efforts are needed to implement the best structure for storing the data in databases. Therefore, to encourage interoperability between heterogeneous subsystems in an intelligent building, a standard message format is needed for event sensing, perceiving and triggering action that fits the requirement of interoperation among subsystems. To explore these questions, we present an integrated middleware approach using SOAP protocol for heterogeneous subsystems in an intelligent building. Subsystems in an intelligent building comprise a number of functionalities that are associated with different events and functions. The need for interoperability in managing these subsystems has led towards transition of vendor independence and open systems. The proposed middleware framework will be able to support multiple services and devices integrated using interoperation table in enterprise database. The interoperation table component is configured as the database of services as well as interfacing with switching module and remote client. The connectivity of the subsystems takes place via Ethernet configuration. Ethernet is ideal for building environment due its performance in real-time as well as taking into consideration cabling of Cat5 that is readily available infrastructure in an intelligent building. The messaging mechanism between subsystems and an intelligent building is based

on SOAP protocol. SOAP is chosen for its interoperability characteristic and defines a standard mechanism of message exchange using XML envelope as payload. The main advantage of SOAP usage is that it provides an open standard for end-to-end communication that is vendor independent as well as a high degree of flexibility for disparate systems integration [14]. SOAP as Web Services technology would enable message exchange between two different subsystems regardless of operating platform or language used. The programming language that was selected for programming the middleware was the C# language with .NET Framework. C# was selected to for its cross-platform support on desktop. .NET Framework was selected here based on its cross-platform independence when running various subsystems with different operating platforms. The system architecture (Fig. 3) shows the entire subsystems configured to achieve interoperability using SOAP and Web Services.

As we know the philosophy of having middleware in an intelligent building is to provide agreement for integration and act as service mediator between diversified subsystems, therefore there must be rules and procedures that could enable subsystems to work together in a federated manner. To realize the middleware's mediator ability, the ECA interoperation schema is proposed. The proposed middleware will perform interoperation among various heterogeneous systems integrated in an intelligent building environment via ECA interoperation schema.

3.1. Building management server

The main component that forms the middleware architecture consists of building management server. The building management server communicates with all the heterogeneous subsystems, and

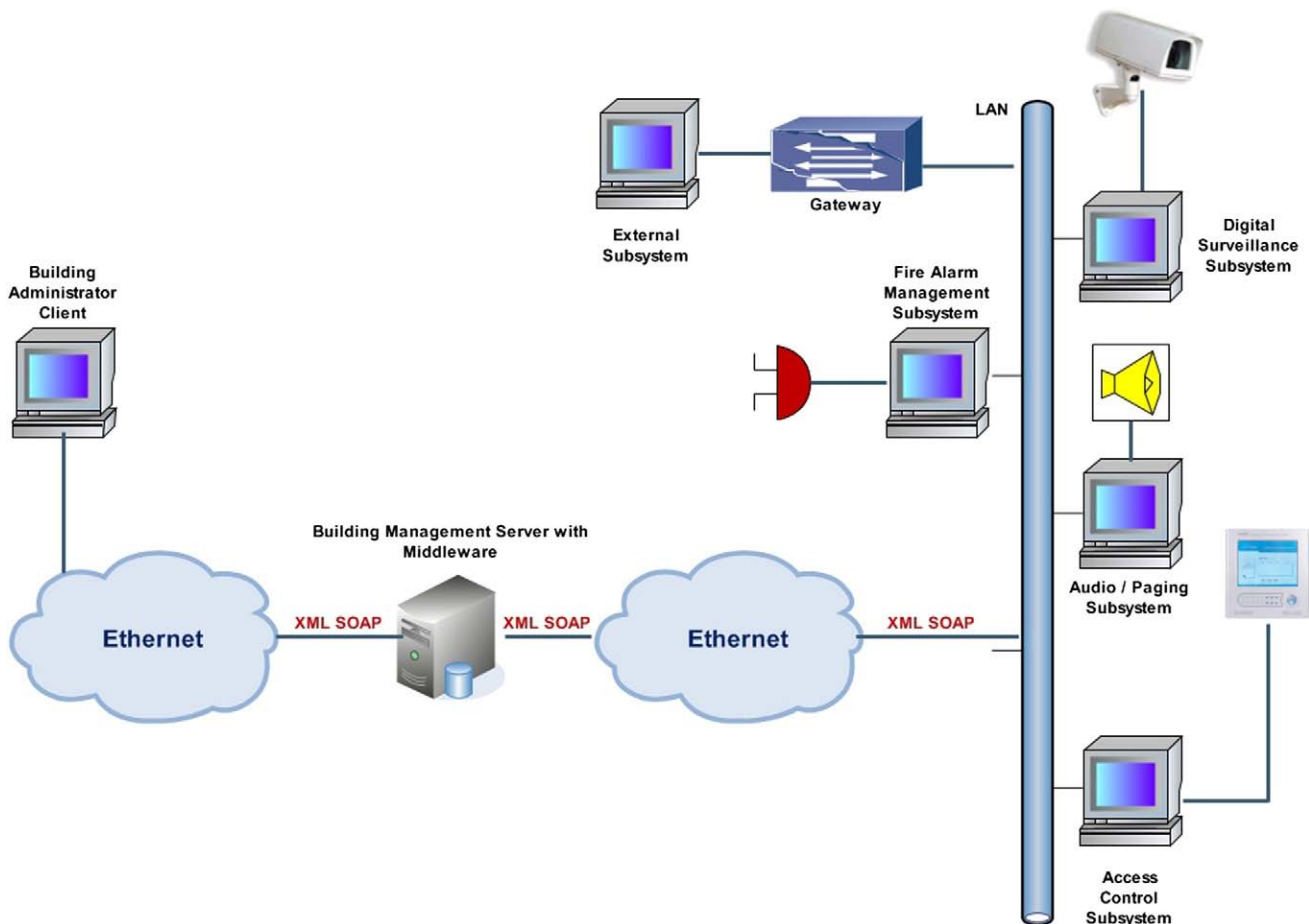


Fig. 3. System architecture.

hosts the drivers and software engine, configured via Ethernet connectivity. The server will perform data retrieval and storage from underlying subsystems, through the database module for control operations of the respective subsystems. The server acts as the storage for hosting third party drivers. The server is configured using Windows Server 2008 and Internet Information Services 7.0. .NET Framework 3.0 [15] is installed and configured together with SQL Server 2008 as the database module in the backend. In addition, the building management server is the entry point for access to each subsystem, acting as a residential gateway, providing the required security and connectivity functionalities.

3.2. Database module

A database module is configured to manage the queries of SOAP messages received from the heterogeneous systems configured in an intelligent building. Microsoft SQL Server is used as a mediator for message exchange and transactions between subsystems. The database module consists of a relational database which exposes methods for the management of the data storage/retrieval procedure. The module stores data from all the subsystems together with their system-centric information such as status, response and action codes. All these data are organized in three-tiered tables named as Status, Response and Action tables. Fig. 4 below shows the database module architecture.

3.3. ECA interoperation schema

Intelligent buildings are event-driven in nature, hence there is a need to normalize unpredictable and interoperation tasks in a building environment. We can realize this by embedding intelligence in the middleware architecture. Therefore, to realize such intelligent interoperation, we propose the ECA interoperation schema. The ECA interoperation schema is based on Event–Condition–Action (ECA) rules. ECA rules were first introduced in the domain of database

technology [26]. ECA rules take the form “on” some event “if” some condition *then* some actions. Here, *events* are defined as a happening of interest at a particular point of time. If *conditions* pertaining to the events are evaluated as true, then one or more *actions* are taken. Once an occurrence of the *event* described in the ON clause takes place, if the *condition* represented in the IF clause is true, the *action* as stated in the DO clause is executed by the system. The rule is said to be triggered when the ON clause is satisfied by the rule and the IF clause is satisfied. Events in the intelligent building can be classified into two categories; internal events and external events. Internal events are events that require a triggering condition by a single event. For example, an access control subsystem will trigger the door access sensor to be opened daily from 8 AM to 5 PM. On the other hand, external events refer to a situation or scenario that requires a triggering condition by two or more events. For example, in the event when a smoke detector detects smoke, the fire management system monitoring will trigger the audio subsystem to page an evacuation message at the affected zones. ECA rule based approach for heterogeneous systems management is advantageous in providing reconfiguration for high-level management and control systems like an intelligent building. ECA rules are considered useful in an intelligent building because: 1) a subsystem needs to react towards an occurrence of some set of events; 2) the reaction of a particular subsystem is subject to some conditions other than standard occurrences of the event; and 3) application logic should be separated from execution of procedures in an intelligent building environment. Due to the constant interaction between subsystems and changes in times, ECA rules seem viable in providing an event-driven mechanism for intelligent building operation. An example of a rule for subsystem interoperation is depicted below:

```
On <FA_alarm_triggered>
If <status_zone001_enable>
Do <trigger_PA001>
```

In the middleware architecture, we specify a set of rules with corresponding actions in a particular subsystem that can be detected and subsequent actions taken based on the requirement defined by the rules. These active rules must be flexible and enable external modification by facility managers or building owners. A rule in the ECA interoperation schema are defined using three-tiered tables named as Status, Response and Action tables. Status table provides data on current status of subsystems, internal events and external events. Response table provides data on subsystems variable and reference codes. Action table provides zone ID, action codes, filename and messages. Each subsystem will have its own action table configured in the middleware. Some exceptions are for fire alarm management system or time scheduler systems as their functions are centered purely on monitoring basis and not into control level. Fig. 5 below shows the structure of the ECA interoperation schema implemented in the middleware.

3.4. Operational scenario

An operational scenario shown in Fig. 6 below shows the sequence diagram on the interoperation of heterogeneous subsystems using SOAP messages.

The operational scenario involves an access control subsystem and digital surveillance subsystem that provides security and monitoring services in an intelligent building. When an intrusion is detected, a triggering signal will be sent to the middleware using a SOAP message by the digital surveillance subsystem. Upon acceptance of the condition, the middleware would need to initialize the associated action defined in the Response table by checking the action codes. The Response table will initialize the Action table to generate the appropriate SOAP message to perform the action sequence resulting

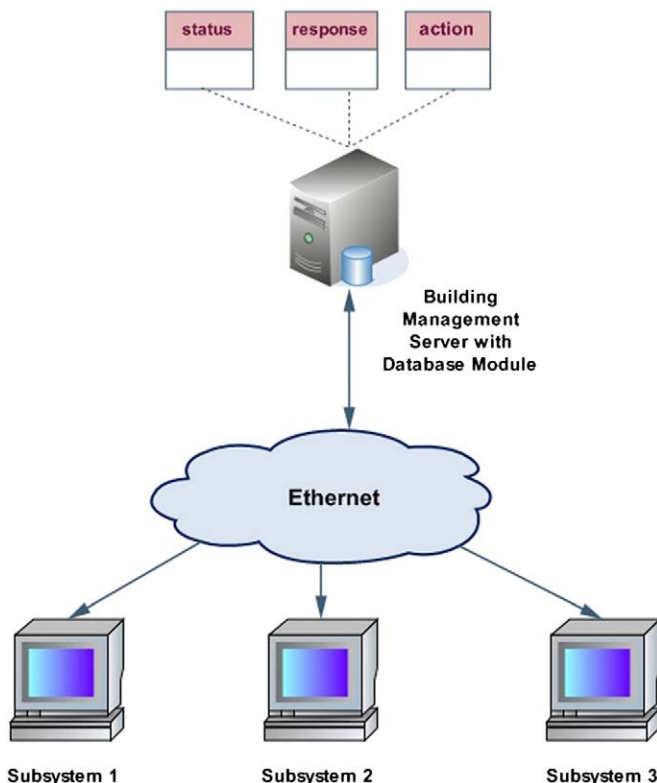


Fig. 4. Database module architecture.

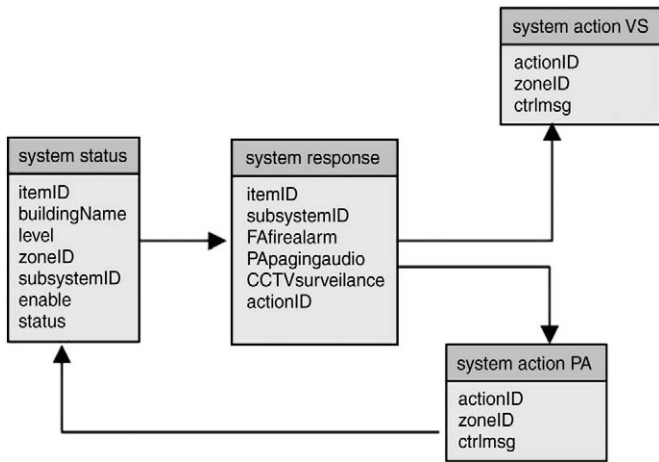


Fig. 5. Structure of the ECA interoperation schema.

of the interoperation requirement. Here, the middleware will ensure that message exchange takes place in a timely manner. After the acceptance of the query, the middleware will send an action message using SOAP to the digital surveillance subsystem or other subsystems that requires interoperation in order to notify security status to building residents. The Status table is updated for every transaction during the interoperation phase. Such scenario clearly requires heterogeneous subsystems to integrate together in a federated approach. Table 2 below shows some of the event categorization identified for middleware operation.

4. Experimental work and evaluation

In this proposed middleware, any SOAP-based compliant subsystem can easily interoperate with each other for building management and control. SOAP gets the desired data types without any complicated data transportation process. During the SOAP message transaction it is interesting to note that message exchanging process are hidden from programmers as they are hidden inside the SOAP middleware. Therefore facility managers can focus on decision making towards data types to be received and the associated interoperation. The data exchange mechanism is done by the SOAP's API provided by the middleware. As long as those third part

Table 2
Event categorization for SOAP-based middleware.

EventID	Event	Condition	Action
1	<from CCTV> guest or visitor walking in Zone 1 that is main entrance	Day time (8.00 AM till 5.00 PM) and Zone 1 is (main entrance)	The middleware will send a signal to trigger audio subsystem (PA) and will stream page welcome message
2	<from Fire alarm> fire alarm triggered indicating emergency	Residents are in the building identified as Zone 2 (meeting room)	Trigger audio subsystems and stream evacuation message referencing the file as coded in Action table.
3	<from Fire Alarm> fire alarm triggered indication emergency	Residents are in the building identified as Zone 3 (office area)	Trigger video subsystem (CCTV) to start recording video file 1 as coded in Action table

subsystems can support standard database system with ODBC-like or SOPA interface, the middleware could maximize the potential of interoperability among heterogeneous subsystems. Three subsystems interoperation has been demonstrated using the SOAP-based middleware in providing interoperability. Those subsystems are loosely coupled or glued by SOAP technology. Therefore, the proposed middleware can be configured to run in any platform or operating systems. The advantages of using SOAP-based middleware are: 1) building application developers can use any programming platform that they are most familiar with; 2) enabling cross-platform deployment; and 3) the service or application software of different subsystems can be upgraded or added independently, as different subsystems are loosely coupled and their dependencies are through the provided middleware and ECA interoperation schema. Since the middleware is based on SOAP technology, it supports a wide choice of programming platform environments. Hence, the middleware can be catered for heterogeneous subsystems in intelligent buildings. The proposed middleware could leverage several existing building systems. Facility managers or building owners have been making investments in intelligent buildings for decades. Today, there are dedicated control networks, controllers, and user interfaces for various building subsystems. Unfortunately, these systems become obsolete as they have dedicated and cumbersome front-end interfaces and little integration to other systems. Using the proposed middleware, legacy systems can support new applications and advanced

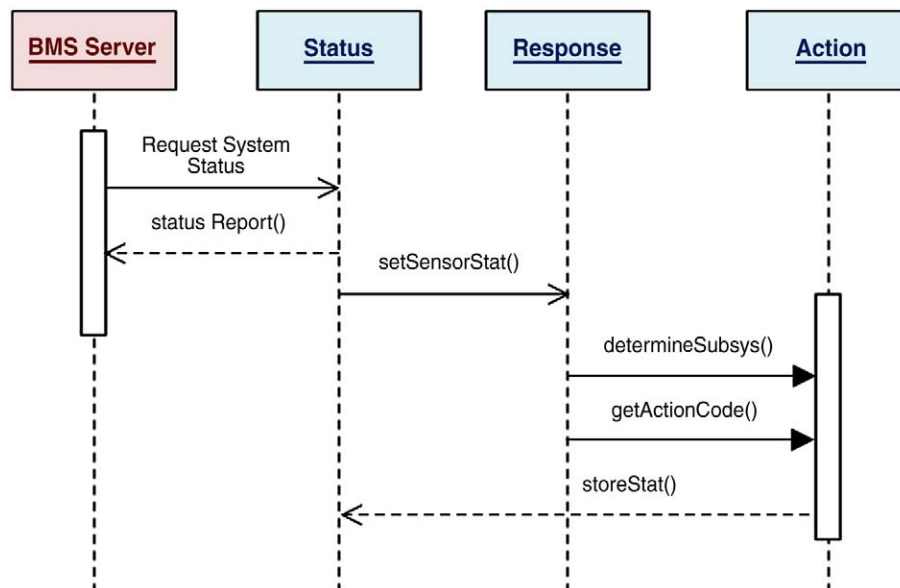


Fig. 6. Sequence diagram for interoperation of heterogeneous subsystems.

features, thereby extending their useful life and reducing capital expenditures. To study the performance of the heterogeneous subsystems interoperation using the SOAP-based middleware, an experimental setup is done to perform an analysis of SOAP message transactions. The testing was carried out to verify the middleware capability in terms of performances governing heterogeneous subsystems operation. Fig. 7 below shows the experimental setup to test the middleware performance.

SOAP protocol is still in its infancy when it comes to cross-platform communication for an intelligent building. Another obvious reason of using SOAP in intelligent building systems is because there is no requirement for intercommunication across firewalls as heterogeneous subsystems are bespoke when it comes to functionalities. For an intelligent building, performance evaluation over network load is important to ensure smooth function of load intensity during concurrent request while operating. The test results are computed with 499 samples for each network load. The results were computed in standard deviation to evaluate the performance of the middleware in each bandwidth requirement. Table 3 below shows the obtained testing results for the interoperation of heterogeneous subsystems.

From Table 3, the testing results show that the standard deviation for performance analysis without load is 2.70. Subsequently, testing with a bandwidth of 8.192 Mbps load gives 3.48 while 16.384 Mbps load gives 5.89 standard deviation values. The rest of the testing results are as follows: 24.576 Mbps load computed with 6.15, 32.768 Mbps load with 7.43 and finally 40.960 Mbps load with 7.54. There is a significant gradual increase in the series of standard deviation for the entire tested network load. Based on the results obtained it can be concluded that there are two significant performance drops of network load with one at 10% and the other one at 30% of network load which are justifiable in the Ethernet environment. This average value obtained also shows that the performance of SOAP in the intelligent building is justified and met the requirement in managing heterogeneous devices.

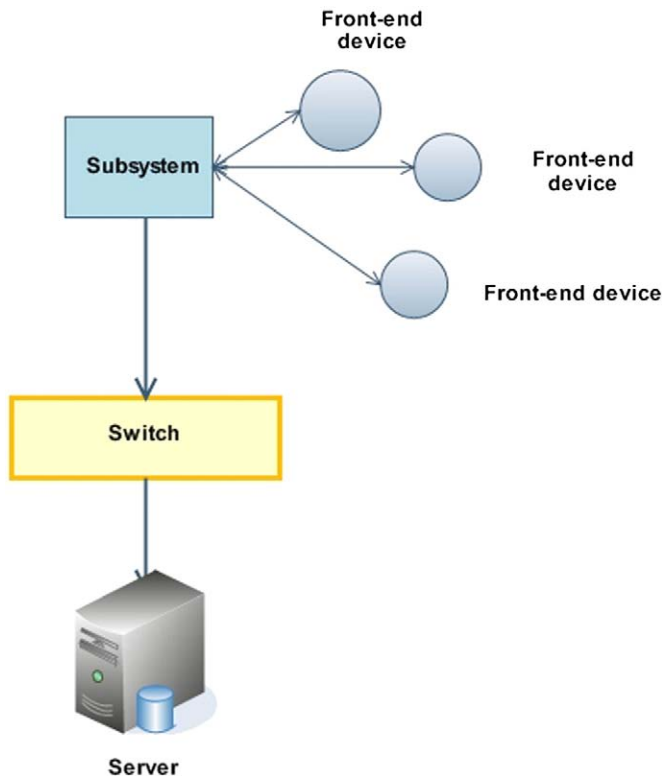


Fig. 7. Experimental design.

Table 3

Middleware performance of network load.

Network load	Standard deviation
No load	2.70
8.192 Mbps load	3.48
16.384 Mbps load	5.89
24.576 Mbps load	6.15
32.768 Mbps load	7.43
40.960 Mbps load	7.54

5. Conclusion and discussion

This paper introduced the SOAP-based middleware, an integrated platform for heterogeneous subsystems management in intelligent buildings. The system is designed in such a way that it caters the flexibility and interoperability of the XML based SOAP protocol in extending the functionalities of subsystems provided by various manufacturers. The developed system is based entirely on the SOAP protocol which overcomes drawbacks of CORBA and other proprietary standards and solutions. SOAP provides scalability and flexibility in ensuring the intelligent building supports existing legacy systems as well as integrating new subsystems or devices. The contribution made by designing this middleware enables the subsystem's logical dependencies tied together in a federated manner. The middleware equipped with both interoperability and coordination ability using ECA interoperation schema enables timely interoperation among heterogeneous subsystems. SOAP implementation would also lead to transparent middleware approach bundled with universal connectivity. In the near future, Web Services and SOAP protocol will have a major impact in providing interoperability for intelligent buildings. An interesting challenge would be to develop a generic abstraction layer based on the defined schema for managing heterogeneous systems with predefined rules extending to heterogeneous subsystems in intelligent buildings.

Acknowledgments

The authors would like to extend their utmost appreciation to the officers in the Institute of Advanced Technology for their support throughout the project period. The authors also would like to thank the Malaysian Ministry of Science, Technology and Innovation (MOSTI) for funding this work under the Science Fund research grant titled Distributed Embedded System of Multimedia Applications for Ubiquitous Home Entertainment (01-01-04-SF0253).

References

- [1] B.M. Flax, Intelligent buildings, *IEEE Communications Magazine* 29 (1991) 24–27.
- [2] J.K.W. Wong, H. Li, S.W. Wang, Intelligent building research: a review, *Automation in Construction* 14 (1) (2005) 143–159.
- [3] P. Davidsson, M. Boman, Distributed monitoring and control of office buildings by embedded agents, *Information Sciences* 171 (2005) 293–307.
- [4] L. Jian, L. Xiaoqin, Z. Xiaoli, Y. Chongchong, Automatic control system of intelligent building based on WebAccess, 7th World Congress on Intelligent Control and Automation, 2008(WCICA 2008), 2008, pp. 7079–7084.
- [5] J. Yang, H. Peng, Decision support to the application of intelligent building technologies, *Renewable Energy* 22 (2001) 67–77.
- [6] J. Duangsuwan, L. Kecheng, Multi-agent control of shared zones in intelligent buildings, *International Conference on Computer Science and Software Engineering*, 2008, pp. 1238–1241.
- [7] S. Wang, Z. Xu, H. Li, J. Hong, W.-z. Shi, Investigation on intelligent building standard communication protocols and application of IT technologies, *Automation in Construction* 13 (5) (2004) 607–619.
- [8] W. Kastner, G. Neugschwandtner, S. Soucek, H.M. Newmann, Communication systems for building automation and control, *Proceedings of the IEEE* 93 (6) (2005) 1178–1203.
- [9] IEEE standard computer dictionary. A compilation of IEEE standard computer glossaries, *IEEE STD 610*, 1991.

- [10] V. Miori, L. Tarrini, M. Manca, G.A.T.G. Tolomei, An open standard solution for domotic interoperability, *IEEE Transactions on Consumer Electronic* 52 (1) (2006) 97–103.
- [11] P. Louridas, SOAP and Web Services, *IEEE Software* 23 (6) (2006) 62–67.
- [12] A.P. Kalogeras, J.V. Gialelis, C.E. Alexakos, M.J. Georgoudakis, S.A. Koubias, Vertical integration of enterprise industrial systems utilizing web services, *IEEE Transactions on Industrial Informatics* 2 (No.2) (2006) 120–128.
- [13] T. Perumal, A.R. Ramli, C.Y. Leong, S. Mansor, K. Samsudin, Interoperability among heterogeneous systems in smart home environment, *IEEE International Conference on Signal Image Technology and Internet Based Systems, (SITIS '08)*, 2008, pp. 177–186.
- [14] M. Neugschwandtner, G. Neugschwandtner, W. Kastner, Web Services in building automation: mapping KNX to oBIX, *5th IEEE International Conference on Industrial Informatics*, 2007, pp. 87–92.
- [15] A.B. Philip, "Middleware: a model for distributed system services", *Communications of the ACM* 39 (2) (1996) 86–98.
- [16] S. Wang, Z. Xu, J. Cao, J. Zhang, A middleware for web service-enabled integration and interoperation of intelligent building systems, *Automation in Construction* 16 (1) (2007) 112–121.
- [17] T. Hannelius, M. Salmenpera, S. Kuikka, Roadmap to adopting OPC UA, *6th IEEE International Conference on Industrial Informatics (INDIN 2008)*, 2008, pp. 756–761.
- [18] S. Chemishkian, J. Lund, Experimental bridge LonWorks/UPnP: 1.0, *First IEEE Consumer Communications and Networking Conference, CCNC*, 2004, pp. 400–405.
- [19] R. Elfving, U. Paulsson, L. Lundberg, Performance of SOAP in Web Service environment compared to CORBA, *Ninth Asia-Pacific Software Engineering Conference*, 2002, pp. 84–93.
- [20] A. George, T. Aphrodite, P. Michael, Interoperability among heterogeneous services, *SCC '06, IEEE International Conference on Services Computing*, 2006, pp. 174–181.
- [21] R.P. Diaz Redondo, A.F. Vilas, M.R. Cabrer, J.J. Pazos Arias, L. Marta Rey, Enhancing residential gateways: OSGi service composition, *IEEE Transactions on Consumer Electronics* 53 (1) (2007) 87–95.
- [22] V. Kapsalis, K. Charatsis, M. Georgoudakis, E. Nikoloutsos, G. Papadopoulos, A SOAP-based system for the provision of e-services, *Computer Standards & Interfaces* 26 (No.6) (2004) 527–541.
- [23] ASHRAE SSPC 135, <http://www.bacnet.org>
- [24] EIA Standard "Control Network protocol specification" EIA/CEA-709.1B (Revision of EIA-709.0-A).
- [25] Y.-M. Hsieh, Y.-C. Hung, A scalable IT infrastructure for automated monitoring systems based on the distributed computing technique using simple object access protocol Web-services, *Automation in Construction* 18 (4) (2009) 424–433.
- [26] R. Adaikkalavan, S. Chakravarthy, "SnoopIB: interval-based event specification and detection for active databases", *Data & Knowledge Engineering* 59 (1) (2006) 139–165.